

RESEARCH ARTICLE

Machine Learning Pipeline for Power Electronics State of Health Assessment and Remaining Useful Life Prediction

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ABSTRACT Power electronic devices are integral to numerous applications in modern technology. The non-linear degradation and subsequent abrupt failure of these devices demand regular State of Health (SoH) checkups as well as accurate prediction of their Remaining Useful Life (RUL). SoH and RUL enable efficient maintenance planning, cost-effective resource allocation, and enhanced safety measures. Consequently, this paper proposes a computationally efficient machine-learning pipeline capable of accurately predicting the SoH and RUL of power electronic devices. To evaluate the effectiveness of the proposed approach, we conduct an exploratory analysis of MOSFET (Metal Oxide Field Effect Transistor) device degradation data. A total of 20 power MOSFETs are subjected to stress tests, mimicking real-world operating conditions. We then carry out pre-processing steps on collected data and train a two-stage machine learning pipeline. In the first stage, a non-parametric model classifies the state of the device as healthy or pre-failure. To save computational time, the second stage consists of a regression model that only triggers, when the device is classified as pre-failure, forecasting its resistance degradation trajectory. The combination of Random Forest classifier and Bayesian Ridge regressor effectively captures the non-linear degradation behaviour of the MOSFETs and accurately estimates SoH with an average accuracy of 80% for the classification step and an average RMSPE of 1.25% for the RUL regression step, respectively. Beyond power electronic devices the proposed pipeline can be adapted to determine the RUL of other electronic devices with non-linear degradation behaviour, thus becoming a methodology for future studies in this domain.

INDEX TERMS Accelerated life cycle test, data-driven prognostics, machine learning, MOSFET, MOSFET RUL, predictive maintenance, power electronics remaining useful life, remaining useful life estimation, state of health estimation.

ABBREVIATIONS

ANN Artificial Neural Network.
AI Artificial Intelligence.
ALT Accelerated Life Test.
BJT Bipolar Junction Transistor.
BRR Bayesian Ridge Regression.

CNN Convolutional Neural Network.
DD Data-driven.
DAQ Data Acquisition Unit.
DUT Device Under Test.
EVs Electrical Vehicles.
EKF Extended Kalman Filter.
FET Field Effect Transistor.
GA Genetic Algorithm.
GaN Gallium Nitride.

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GBMs	Gradient Boosting Machines.
HVDC	High-voltage Direct-current.
HCI	Hot Carrier Injection.
IGBT	Insulated Gate Bipolar Transistor.
KF	Kalman Filter.
LS	Least Square.
LSTM	Long Short-Term Memory.
MCC	Matthews correlation coefficient.
ML	Machine Learning.
MOSFET	Metal Oxide Semiconductor Field Effect Transistor.
NI	National Instrument.
PF	Particle Filter.
PHM	Prognostics and Health Management.
RF	Random Forest.
RANSAC	Random Sample Consensus.
RVM	Relevance Vector Machine.
RMSE	Root Mean Squared Error.
RMSPE	Root Mean Squared Percentage Error.
RNN	Recurrent Neural Network.
RUL	Remaining Useful Life.
Si	Silicon.
SiC	Silicon Carbide.
SOH	State of Health.
SVM	Support Vector Machine.
TDDDB	Time-dependent Dielectric Breakdown.
WBG	Wide Band Gap.

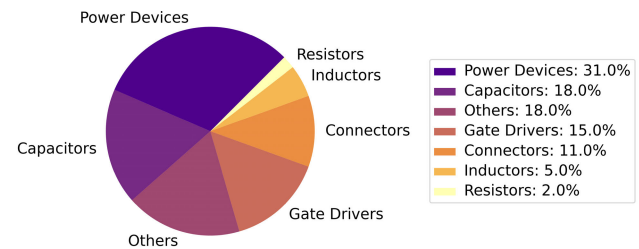


FIGURE 1. Failure rates of electronic components (data retrieved from [4]).

into two categories: wear-out failures and catastrophic failures. In general, wear-out failure is the outcome of a component's cumulative degradation over time. Hence, it may be prevented by condition monitoring. Condition monitoring is a framework that aims to collect data on critical equipment regarding its condition for fault-free operation [5]. On the other hand, chip-related failures such as over-current, and over-voltage are considered as catastrophic since these failures are instantaneous [6]. In this work, we concentrate on cumulative degradation failure as such failures can be prevented by condition monitoring.

Traditional maintenance and engineering practices are based on corrective actions where costly replacement and hardware redundancy are adopted. However, since there is a fast transition toward high power density and more reliable applications, it is impractical to raise the size and cost of converters (by redundancy of components) to improve their reliability. Variations in manufacturing processes, materials, or environmental conditions during production can introduce uncertainties or defects that affect the reliability and durability of the manufactured products. As a consequence, the traditional threshold-based detection methods fail to capture the device's state of health (SoH). Hence, to be able to monitor the device's health, reduce and/or prevent catastrophic failures, Prognostics and Health Management (PHM) techniques can be adopted [7]. PHM methods include physics-based, data-driven, and hybrid which combines the previous two methods. Considering the increased system complexity, data-driven - PHM can enable useful information regarding device health using historical and online condition monitoring data [8]. However, the use of data-driven-PHM in the power semiconductor components is still under investigation. In most instances, degradation data for the whole life of the component is unavailable due to the complex mission profile of devices. Recall, power semiconductors exhibit one of the highest failure rates in power electronic circuitry. Consequently, there is a need for methodologies that may be utilized to identify the health conditions and forecast the Remaining Useful Life (RUL) of power semiconductors in these situations. In this regard, power semiconductors may be created with a current sense transistor [9] in which current can be monitored with high confidence and low-loss manner enabling data-driven-PHM

I. INTRODUCTION

Modern power electronic systems consist mostly of three types of components: active components (i.e., power semiconductor devices), passive components (e.g., capacitors and inductors), and control units. Power semiconductors (IGBTs and MOSFETs) are one of the most critical parts of modern power electronic circuitry. These devices are being utilized in a plethora of industries, such as automotive, aerospace, energy transmission, conversion, and storage (including battery stationary storage), avionics, rail traction, radar systems, consumer electronics devices, and numerous others [1]. By 2028, the global market for power semiconductors is projected to reach \$52.7 billion, an increase from \$41.4 billion in 2022 [2].

The advancement in power electronics is an outcome of the advancement of power semiconductor technology. However, during their operational lifetime, power semiconductors might be subjected to significant stresses. These excessive stresses can activate various aging processes that later result in catastrophic failures. In comparison to other components in the electronic circuitry, power semiconductors and capacitors are among the least reliable parts of an electronic device or are the components with the highest failure rate [3]. Based on a survey [4], the failure rates of the various components of converters are categorized as illustrated in Figure 1. Power semiconductor device failures can be divided

solutions for high reliability-required and mission-critical applications, for instance, automotive, aerospace and satellite applications.

This paper introduces a Machine Learning (ML) pipeline approach capable of accurately estimating the SoH and the RUL of power electronic devices. Crucially, once trained, our ML pipeline can estimate the SoH of power electronics in real-time without any historical data irrespective of manufacturing variations or components (semiconductor devices, capacitors, etc.). To evaluate the effectiveness of this approach, an exploratory analysis is conducted on degradation data obtained from the Accelerated Life-cycle test (ALT) of power MOSFETs. A total of 20 Si power MOSFETs are subjected to controlled stress tests, simulating real-world operating conditions. The collected data undergoes pre-processing steps, and a two-stage ML pipeline is trained.

In the first stage of the pipeline, a never-before-used non-parametric model is employed to classify the state of the device as either healthy or in a pre-failure condition. This classification step plays a critical role in determining when the device requires attention or maintenance. In the second stage, a regression model is utilized to forecast the resistance degradation trajectory of the MOSFET once it is classified as pre-failure by the previous step. The regression model enables accurate prediction of the RUL, a crucial parameter for effective maintenance planning. To capture the non-linear degradation behavior exhibited by MOSFETs, the machine-learning pipeline combines the Random Forest (RF) classifier and the Bayesian Ridge regression (BRR). The integration of these models proves to be highly effective, as evidenced by an average accuracy of 80.7% in the classification step and an average RMSPE of 1.25% in the regression step. These results demonstrate the pipeline's ability to accurately predict the RUL of MOSFETs. The selection of these models is based on their distinct advantages:

- **Random Forest:** Ability to generate models quickly and make predictions rapidly, which makes it computationally efficient. RF is robust to overfitting and can effectively manage a large number of input variables. This characteristic is advantageous when classifying intricate deterioration patterns in MOSFETs [10].
- **Bayesian Ridge Regressor:** Provides probabilistic forecasts and integrates previous information, which is valuable for regression problems that include noisy and uncertain data. Utilizing BRR enhances the accuracy of predicting RUL.

It is important to emphasize that the suggested framework is not restricted to RF and BRR. To demonstrate the effectiveness of the proposed framework, we present results obtained from various algorithms, including Logistic Regression (LGR), Support Vector Classifiers (SVC), and the Random Sample Consensus (RANSAC) Regressor. The pipeline may be modified to include more widely used machine learning models based on the unique needs of the application and the characteristics of the data. We have

demonstrated that the proposed pipeline works with various algorithms by adding two more algorithms for comparison in both classification (namely LGR and SVC) and regression (RANSAC and 2nd order BRR) estimations.

In addition to the ML pipeline, this research also conducted a failure analysis of the power MOSFETs to gain valuable insights into the underlying degradation mechanisms and failure modes. This analysis contributes to a deeper understanding of the degradation processes and aids in the development of more accurate SOH and RUL prediction models. It is important to note that, in this study, the datasets used were created specifically based on real-life operation profiles of MOSFET devices, where devices undergo several temperature cycles under biased conditions. By creating and sharing these datasets, we also contribute to the research community by providing a valuable resource for further exploration and validation of the proposed machine-learning pipeline. Researchers and practitioners can leverage these datasets to replicate and build upon the study's findings, enhancing the transparency and reproducibility of the research.

The paper is organized as follows. Section II - *MOSFET device background* provides a brief MOSFET background, and reviews the failure mechanisms and RUL prediction methods of power MOSFETs. Section III - *Machine learning pipeline approach* introduces the proposed RUL estimation model methodology. The details of data collection and experimental data analysis are given in Section IV - *Data Collection* and Section V - *Experimental data analysis*, respectively. Results of the proposed RUL pipeline are given in Section VI - *Results*. The applicability of the proposed method is discussed in Section VII - *Discussion on practical applicability of the pipeline*. Finally, Section VIII - *Conclusion* summarizes the primary findings and outlines directions for future research.

II. MOSFET DEVICE BACKGROUND

A. POWER MOSFETS BASICS

A MOSFET typically has three regions, namely *drain*, *source*, and *gate*. In the industry, different types of MOSFET technologies are developed e.g., in *planar* and *trench* structures, *N-type* and *P-type*, *enhancement* and *depletion* types. One key parameter to assess a MOSFET is its on-state resistance - that is the total resistance seen between the *drain* and *source* terminals ($R_{ds(on)}$) when the gate is biased. Once a sufficient voltage is applied across the *gate-source* terminals, the current flow starts between *drain-source* electrodes [11]. Due to the intrinsic properties of semiconductors, this current is limited by the on-state resistance and hence by the heat produced. The on-state resistance is the sum of several individual resistances as shown in Figure 15 in Appendix.

B. FAILURE MECHANISMS AND PRECURSORS OF MOSFETS

Since power MOSFETs are used in various applications in different temperatures, loads, frequencies, and environments,

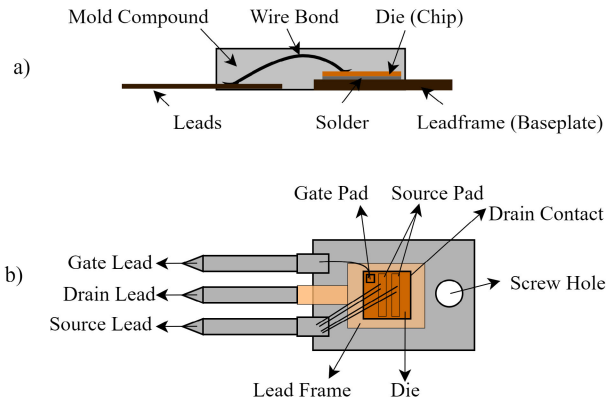


FIGURE 2. a) The TO-247 MOSFET cross-section view with different layers b) The top view of the TO-247 MOSFET.

an understanding of the failure mechanism is crucial to provide a reliable and safe operation during the lifetime. The failure mechanisms of power MOSFET can be distinguished into two groups: extrinsic (package-related) and intrinsic (die/chip/semiconductor-related) failures [1]. A great deal of intrinsic failures—and die-related failures in particular—are due to issues during the manufacturing process causing defects. These failures are usually eliminated during the manufacturing phase before being delivered to end-users [12]. Therefore, the intrinsic failures are not in the scope of this paper. Extrinsic failures are related to packaging and/or wear-out failures. The packaging-related failure mechanisms are related to thermo-mechanical stresses. These are the most prevalent failure mechanisms arising after long usage periods, or extensive exposure to environmental stress factors such as moisture, vibration, and elevated temperatures. These types of failures are observed more frequently in power electronics compared to intrinsic failures by means of performance parameter deviations e.g., voltage, current, etc. Therefore, the scope of this paper focuses on cumulative degradation mode e.g., packaging-related failures. A cross-section view of the frequently used MOSFET device with the TO-247 package is presented in Figure 2. As can be seen from the illustration, the device has different layers and components utilized during manufacturing. Therefore, overall device and package reliability strongly depends on the robustness of its individual layers. Even while certain innovative packaging technologies have been developed specifically for power modules to increase reliability, the bulk of commercial MOSFET devices still use traditional packaging and wire bonding methods [13]. As detailed above, the device characteristics of a MOSFET are prone to alter over time. These changes can be measured through some of the electrical device parameters. Therefore, it is important to distinguish changes in these parameters for pinpointing the failure location and understanding the failure mechanisms behind it. Failure precursors are indicator parameters associated with a possible incoming failure event. In the literature, an increase in $R_{ds(on)}$ is the most popular failure precursor exploited to assess package related

MOSFETs degradation and/or failures due to weakening bond-wire and solder fatigue [14], [15], [16], [17]. The $R_{DS(on)}$ value increases non-linearly with respect to the degradation of a power MOSFET [15], [18]. In the literature, the 5% of increase in on-state resistance is regarded as a failure [14], [16], [19]. However, it is important to highlight that this threshold is chosen for the sake of safety. In addition, it is emphasized that after a 5% increase in $R_{DS(on)}$, the device remains operational until its complete destruction. Similarly, the increase in *source to drain* (V_{SD} , the forward voltage drop of the integral body diode) is another packaging-related failure precursor and its deviation provides information on the die attach health, and the thermal behaviour of the device [17]. On the other hand, the chip gate oxide-related failures such as Time-dependent Dielectric Breakdown (TDDb) and Hot Carrier Injection (HCI) are mostly identified by means of I_{gss} , V_{th} and *miller plateau shift* [16], [20], [21], [22]. Even though different degradation/failure precursors have been defined above, it is essential to note that every degradation process has the potential to either cause or result in a set of degradation mechanisms.

C. REVIEW OF DATA-DRIVEN RUL PREDICTION METHODS OF POWER MOSFETS

ALT tests are conducted to collect device parameter data in less time. Failure precursors can be extracted from these data to understand the failure mechanism of a device. The development in ML and Artificial Intelligence (AI) further encourages the use of data-driven maintenance methods in mission-critical applications. Employing data-driven maintenance techniques to power MOSFETs can improve overall system reliability. As described, the device characteristics of MOSFET are influenced by a variety of factors, such as temperature, vibration, humidity, voltage, etc., and change over time. In recent years, numerous studies have been conducted to extract data from sensor data and employ it with data-driven approaches to prevent and/or mitigate such failures [18], [23], [24].

In the literature, many different algorithms, such as Kalman Filter (KF) [23], [25], [26], Particle Filter (PF) [18], [26], [27], [28], Relevance Vector Machine (RVM) [29], Least Square (LS) [14], [15], [30] etc., were used to predict MOSFET RUL. A comprehensive overview of data-driven prognostics for power MOSFETs is presented in Table 1.

Among the aforementioned failure precursors, $\Delta R_{ds(on)}$ is mostly used due to its effectiveness and applicability to the real world. In [19], a dedicated power cycling test setup was utilized to collect device data. Then, in order to obtain a linear increase in $\Delta R_{ds(on)}$, natural logarithm ($\ln(\Delta R_{ds(on)})$) is used. Further, the Random Sample Consensus (RANSAC) algorithm has been employed to remove outliers from the linearized $\Delta R_{ds(on)}$ function. The RUL estimation is made using classical-LS and Genetic Algorithm (GA), but it cannot properly capture the non-linearities of the MOSFETs during

TABLE 1. The summary of recent studies on the state of remaining useful life for power MOSFETs.

Ref.	Method ¹																Feature inputs					RUL												
	LS	LN-LS	HA	LN-HA	GA	GP	PF	IPF	APF	KF	EKF	RVM	SVM	LSTM	MLE	MAP	GPR	SKRR	BO	SSA	ESN	RANSAC	MHA	Cycle no.	Time	Drain Voltage	Drain Current	Gate Threshold Voltage	Temperature	$R_{ds(on)}$	V_{th}	Miller Plateau		
[31]									⊗									⊗						⊗		⊗								
[25]	⊗									⊗			⊗												⊗		⊗							
[32]						⊗																							⊗					
[33]													⊗						⊗						⊗									
[34]																			⊗						⊗									
[30]	⊗	⊗	⊗	⊗																	⊗				⊗									
[35]																				⊗					⊗									
[28]							⊗														⊗				⊗									
[19], [36]	⊗			⊗																	⊗				⊗									
[23]	⊗									⊗															⊗									
[15]	⊗																					⊗						⊗						
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[29]												⊗													⊗									
[24]						⊗																						⊗						
[14]	⊗											⊗													⊗									
[18]								⊗				⊗					⊗								⊗									

¹ LS: Least Square, LN-LS: Linearized LS, HA: Horizontal Average, LN-HA: Linearized HA, GA: Genetic Algorithm, GP: Genetic Programming, PF: Particle Filter, IPF: Improved PF, APF: Assisted PF, KF: Kalman Filter, EKF: Extended KF, RVM: Relevance Vector Machine, SVM: Support Vector Machine, LSTM: Long Short Term Memory, MLE: Maximum Likelihood Estimation, MAP: Maximum a Posterior, GPR: Gaussian Process Regression, SKRR: Sparse Kernel Ridge Regression, BO: Bayesian optimization, SSA: State-Space Algorithm, ESN: Echo State Network, RANSAC: Random sample consensus method, MHA: Metropolis-Hastings Algorithm

the early stage. In [14], the estimated model parameters are obtained by applying the LS method, and the RUL is predicted using Extended Kalman Filter (EKF). However, the EKF estimations exhibit a high error. In [34], to perform RUL estimation, a Long Short-Term Memory (LSTM) is used due to its better performance on time-series data than Recurrent Neural Network (RNN). This work showed better performance on the same data set compared to [18]. In [25], LS-Support Vector Machine (SVM) and KF are each used for RUL prediction. The comparison of these two methods shows that the KF outperforms the LS-SVM in terms of stability and accuracy. Although the errors are very low for both KF and LS-SVM models, these models are tested only on one device. Therefore, it is not realistic to employ this model for real-world applications without further confirmation on different devices. Similarly, in [33], the data-set from [18] is used. An LSTM-RNN model is developed to predict the RUL of a MOSFET using $\Delta R_{DS(on)}$. This study shows improved results with low RMSE losses.

Apart from $\Delta R_{DS(on)}$, other failure precursors have also been utilised to estimate the RUL of a device. In [15], different parametric features are investigated to choose the proper failure indicator. It is pointed out that V_{th} is a good precursor candidate to detect gate oxide degradation. The collected data from this investigation is then further used to estimate RUL using RANSAC and LS methods. It is shown that the RUL of three devices can be predicted with high accuracy despite the high variance observed during the early life of the MOSFET. In [21], V_{th} instabilities of the MOSFETs are selected as failure indicators. Yet, it is important to highlight that measuring the V_{th} within real-world applications for the sake of RUL estimation is difficult, if not impossible.

III. MACHINE LEARNING PIPELINE APPROACH

This work adopts a pipeline-based approach, where features are engineered from monitored multivariate time series signals, i.e. voltage and current. Extracted features are used to train a sequential, two-step model that comprises two stages: classification and regression. In the classification stage, a non-parametric model classifies the device's state as healthy or pre-failure. The regression stage employs a model to forecast the resistance degradation trajectory and predict RUL once the classifier predicts that the device reached the pre-failure threshold. Figure 3 illustrates the overall architecture of the proposed ML pipeline whilst this subsection first introduces the data pre-processing and feature engineering steps in Section III-A1 and Section III-A2 respectively, followed by the classification in Section III-B and regression algorithms in Section III-C respectively, developing on key data science methods adopted as required.

A. DATA PREPARATION

1) PRE-PROCESSING

To be able to estimate the MOSFET's SoH and later predict its remaining useful life using the proposed MOSFET

SoH-RUL estimation method, various signals are monitored during MOSFET operation. The monitored signals are voltage, current, and case temperature of the MOSFET device. Drain-source and gate-source voltages, as well as drain current, are recorded during the MOSFET's turn-on sequence as illustrated in Figure 4. In addition, the device case temperature is measured and recorded continuously during the turn-on and turn-off phases of the device.

The collected data from the ALT (see Section IV for a detailed explanation of data collection) is thus pre-processed such that features can be extracted to be later used as input to the training algorithm. The data pre-processing steps adopted in the current work consist of removing all signal data once the device resistance reaches a 7% increase from its initial value as this data is redundant since it is collected after component failure. First, each signal (voltage and current) is divided into equal windows of $n = 10,001$ entries (see Figure 5 for raw signal data and window length n). The large n is empirically chosen here to achieve a high curve soothing, however, smaller/higher values can be used depending on the application. Second, each window's mean value between the 10th (q10) and 90th (q90) percentiles is then calculated to avoid any extreme outliers in each window. Next, the resistance vector is calculated according to Equation (1). Finally, all data beyond a 7% increase in the mean $R_{DS(on)}$ resistance value is removed since it is irrelevant once the device reaches end-of-life (EoL). The obtained degradation curves after data removal can be visualised in Figure 6.

2) FEATURE ENGINEERING

To understand the underlying degradation process of DUT and capture a possible failure precursor, a feature engineering process is applied to the collected and pre-processed data. Statistical features are extracted from a moving window of size $n = 17$ applied on the raw data (see Figure 5). The moving window size was deliberately chosen to be small enough to provide high granularity when extracting features. The mean, standard deviation, skewness, and kurtosis of the drain-source voltages (V_{DS}), and drain current (I_D) values are computed through the time-domain statistical parameters given in Table 8 in the Appendices. $R_{DS(on)}$ per window is also computed using Ohm's law, as shown in eq. (1), and the same features—standard deviation, skewness, and kurtosis are then extracted.

$$R_{DS(on)} = \frac{V_{DS}}{I_D} \quad (1)$$

where $R_{DS(on)}$ is on-state resistance between drain and source terminals, voltage drop and flowing current between drain and source terminals are defined as V_{DS} and I_D , respectively.

In addition to the engineered features, we also record the initial resistance of the device, the percentage increase in resistance as the device is ageing as well the time in seconds (unless otherwise explicitly stated). Note that the three aforementioned recordings are not used as input to the algorithms; rather, they are used for computing statistics in Section VI. The correlation matrix of the extracted features

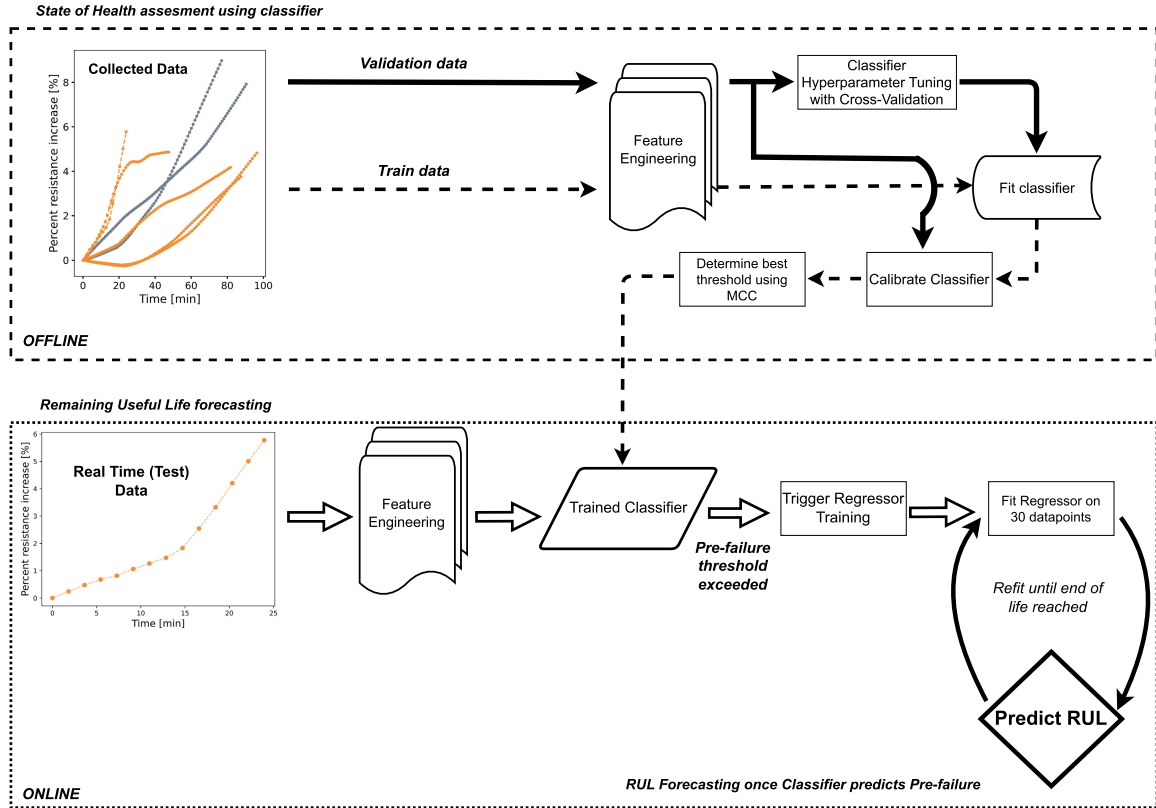


FIGURE 3. The proposed RUL prediction method pipeline.

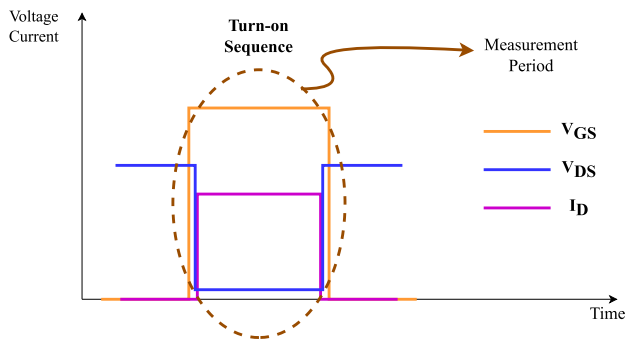


FIGURE 4. Signal extraction during the turn-on of the MOSFET.

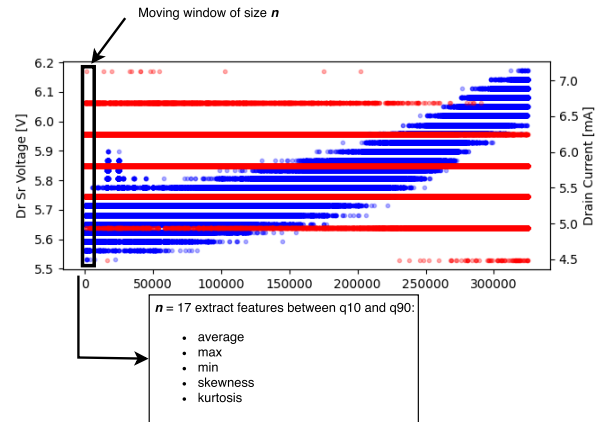


FIGURE 5. Raw signal data (current and voltage) vs time in seconds and moving window of size n . Note: We use $n = 10,001$ for data pre-processing and $n = 17$ for feature extraction. q_{10} and q_{90} represent the 10th and 90th quartiles of the window.

and labels is provided in the Appendix in Figure 18 and Figure 19.

B. STATE OF HEALTH ESTIMATION

From a machine learning perspective, the task of estimating MOSFET health can be regarded as a supervised classification problem. The classifier's task is to categorise the device's state as healthy or pre-failure based on a set of inputs. In this work, we refer to the state of health as defined within the broader context of research, specifically PHM. In PHM, the 'state of health' refers to a comprehensive measure of the condition and performance of a system,

component, or equipment relative to its ideal or expected operating condition. It is a key indicator used to assess the current health status, predict future performance, and estimate the remaining useful life of the asset [39], [40]. To avoid confusion with the term 'State of Health' as used in the Li-ion battery health prognosis field, where SoH typically refers to the remaining usable capacity of a battery,

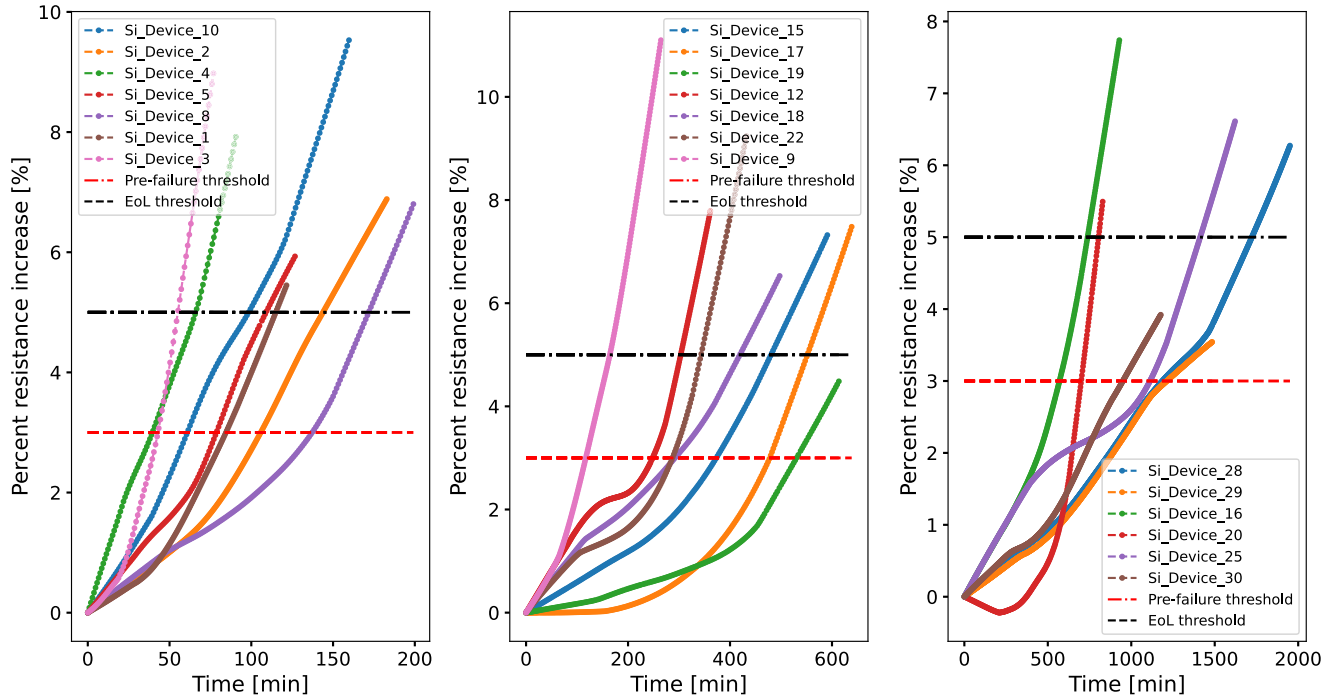


FIGURE 6. Filtered resistance increases with time for all Si MOSFETs that have undergone accelerated life cycle testing and associated pre-failure and EoL thresholds. Note: for visualization purposes, the data has been (1) filtered using a Lowless Filter [38] and (2) split into three figures based on sample survival time.

we explicitly define SoH for MOSFETs in our context as an indicator that quantifies the device's health status based on changes in its electrical parameters, particularly the on-state resistance, $R_{DS(on)}$. We determine whether a device is healthy (labelled as 0) or pre-failure (labelled as 1) based on a pre-failure threshold. We call the label as warning delta $R_{DS(on)}$ (refer to the correlation matrix in Appendix Figure 18). We choose a 3% increase as the pre-failure threshold as this represents 60% of the device's life when the EoL is considered as a 5% increase in $R_{DS(on)}$. It is noteworthy to mention that the thresholds can be adjusted based on the intended application (see Figure 6 for the pre-failure and EoL thresholds). To perform the binary classification of health (0) and pre-failure (1), this work makes use of a Random Forest (RF) classifier [41] implemented in scikit-learn [42]. The random forest algorithm combines multiple decision trees to make predictions. Each decision tree in the random forest is trained on a different subset of the training data and uses a random subset of features for determining the best split at each node. This randomness helps to reduce over-fitting and improve the generalization ability of the model. To make a prediction using an RF classifier, each decision tree in the forest independently predicts the class label and the final prediction is determined by averaging the individual predictions. In this work we do not make use of hard predictions, instead we let the RF predict the probability of each sample belonging to a class. To train the algorithm we first split the data into train, test, and validation (calibration). We then use the validation data containing a total of 8 samples to perform hyper-parameter tuning of the RF using Optuna

optimization framework [43] with a *TPESampler* and a *median pruner* as a single-objective optimization. During the hyper-parameter search, we use the negative log loss to compute the scoring and perform group-based cross-validation to obtain the aforementioned score value - where each group represents a single device sample i.e. train on 7 samples and predict on 1 until the cross-validator runs out of samples. Once the hyper-parameters are determined we then proceed to model calibration. Model calibration refers to the process of adjusting the predicted probabilities or scores generated by the classifier (RF in this case) to align them with the true label. In this case, we want the RF to predict a high probability for samples in a pre-failure state and a low probability in a healthy state as we want to identify when the device is in a pre-failure state. We perform calibration using sigmoid calibration, also known as Platt scaling or Platt calibration [44]. We use the sigmoid calibration method that fits a logistic regression model to the predicted probabilities of the RF model using the validation dataset of 8 samples. We perform the calibration using the validation data so that we do not bias the calibrator. Platt scaling works by fitting a logistic regression model that maps the predicted probabilities to a calibrated probability by applying a sigmoid function. This function has the mathematical form:

$$P_{calibrated} = \frac{1}{1 + e^{(A \times P_{RF} + B)}} \quad (2)$$

where the $P_{calibrated}$ represents the calibrated probability, A , B are the coefficients learned during the fitting of the logistic regression curve, and P_{RF} is the raw probability predicted by

the random forest indicating the likelihood of belonging to the positive class.

Finally, after recalibration, we determine the optimum probability decision threshold using Matthews correlation coefficient (MCC) as per Equation (9). In this last step, we first calculate the MCC for different thresholds from 0.1 to 0.9 and return the best threshold that maximizes the MCC score for our binary classification based on predicted probabilities and true labels. The threshold is then used to convert the predicted probabilities into binary predictions.

A detailed overview of the entire training and testing procedure for the classifier pipeline is visually described in Figure 3. The RF model performance on test data is described in detail in the results Section VI-A.

C. REMAINING USEFUL LIFE FORECASTING

In this work, estimating power MOSFET RUL is also regarded as a supervised learning problem. Unlike previous work using the natural logarithm of $\Delta R_{DS(on)}$ [36], we use the unmodified average resistance ($\Delta R_{DS(on)}$) as the device health indicator because for power MOSFET on-state resistance increases exponentially with respect to device degradation. As discussed earlier we calculate the instantaneous resistance per window (see Figure 5 for the window illustration) as per Equation (1) and we then average each instantaneous resistance value in the window to obtain the $\Delta R_{DS(on)}$ vector for each device (see Figure 6 for an illustration of the $\Delta R_{DS(on)}$ for each device). We define the component EoL as a 5% increase in the $\Delta R_{DS(on)}$ resistance.

In Figure 6 we observe that degradation curves exhibit a predominantly linear behaviour after the chosen pre-failure threshold and non-linear before the aforementioned threshold. For this reason, we allow the highly capable RF algorithm to capture the non-linear degradation behaviour early in the device's life and then trigger the RUL forecasting only when the classifier introduced in Section III-B starts predicting that the device entered the pre-failure state of health (refer to Section VI for an illustration of the RUL forecasting algorithm trigger point). Unlike previous work [19], which makes use of the logarithmic value of average resistance ($\ln(\Delta R_{DS(on)})$), we eliminate this step by using the classifier to track degradation and trigger the regressor during the linear part. In this work, the RUL forecasting algorithms used are RANSAC and BRR regressors. BRR is an algorithm that combines Bayesian inference with Ridge Regression [45]. It estimates the posterior distribution of the regression coefficients, incorporating prior knowledge or beliefs about the coefficients and providing uncertainty measures for more robust and informative inference. The key advantage of BRR is its ability to handle collinearity or multi-collinearity issues by automatically adapting the model complexity. It achieves this by introducing a regularization term (similar to Ridge Regression) that helps control the magnitude of the coefficients. The regularization term helps prevent over-fitting

and improves the stability of the model by shrinking less informative coefficients towards zero. BRR also outputs model confidence by estimating the standard deviation of its prediction. A desirable property when forecasting the degradation of an electronic component. We use the predicted standard deviation to estimate the 90% confidence using Equation (3) with a standard score of $z = 1.645$.

$$CI = \hat{y} \pm z \times \sigma \quad (3)$$

where CI is the confidence interval, z is the standard score, and σ is the standard deviation.

The BRR algorithm is fitted on the resistance ($\Delta R_{DS(on)}$) versus time curve on a total of 30 data points for each of the devices under test. The 30 historical data points for BRR training used are selected once the RF classifier starts predicting that the device entered the pre-failure state. The BRR algorithm is retrained on the fly as newly recorded data of resistance versus time is recorded. For the purpose of this work, we divide the time to EoL into four prediction intervals as follows intervals 100% to EoL (just after the trigger point of the regressor), and 75%, 50% and 25% to EoL respectively. We therefore only re-train the regressor at the end of each prediction interval, however, the regressor can be re-trained on the fly as frequently as desired by the user.

A detailed overview of the entire training and testing procedure for the regressor is visually described in Figure 3. The BRR model performance on test data is described in detail in the results Section VI-B.

IV. DATA COLLECTION

The absence of pertinent lifespan data is a significant barrier to proving the benefits of a prognostics solution in many electronic devices as well as power MOSFETs. An accelerated life cycle platform based on low-voltage, high-temperature power cycling has been created as part of this research effort to address this shortcoming. This test setup aims to replicate real-life operations where devices undergo repeated temperature cycles. For that reason, a reliable and continuously monitorable experimental design, hardware, and software were prepared. As stated in the previous chapter, packaging-related failures are the most prominent failures for power MOSFET applications [12]. The on-state resistance of the power MOSFET is impacted by the packaging-related degradation during the useful lifespan. Therefore, the focus of the proposed experimental test setup is to extract the respective device data.

A. ACCELERATED LIFE-CYCLE TEST BENCH

To be able to record and monitor the experiments, various hardware are required. Figure 16 in the Appendix shows the laboratory test platform of the experiment. As can be seen from Figure 16, a DUT test board has been integrated into the test setup to conduct the ALTs along with an adjustable load resistor, infrared temperature sensor, and forced air fan. There are also two DC power supplies for switching the gate terminal of DUT and biasing the drain

terminal of DUT along with the protection relay which is automatically programmed using NI LabVIEW. On this board, a protective relay is utilized to halt the system if the DUT loses its capacity to switch off. Apart from that, in order to measure the drain current, a current sensor having a nominal current of up to 15A is placed on the board. This board enables the ALT experimentation with drain currents of DUT up to 20A using TO-220 packaged Si power MOSFETs. The National Instrument (NI) cDAQ-9178 controller is the primary component of the experimental arrangement for monitoring and recording the data. This module controls the data acquisition and switching of the tested MOSFET devices. In order to log data and monitor the experiment, the DAQ is connected by Ethernet to a Control Unit (Control-PC). Table 7 in Appendices summarizes the hardware and power MOSFET devices used for conducting the ALT experiment.

1) DESIGN OF ACCELERATED LIFE TEST PLANS

The extent of the temperature fluctuation and changes are the primary causes of package-related failures. To observe the effects of electro-thermal stresses in a shorter amount of time, low-voltage high-temperature accelerated power cycling tests are conducted on purpose-built test beds. The ALT's goal is to repeatedly stress the device with a drain current less than the nominal value while self-heating, that is, without a heat sink. During each cycle, the objective is to induce heat diffusion through all of the device's layers. This heat diffusion is expected to activate the package-related failure modes i.e., delamination, solder voids, metallization, and wire lift-off. Therefore, the DUT is actively heated through switching and conduction, causing power losses within the device, and is cooled down via a cooling air fan once the pre-configured maximum allowable temperature has been reached. This cycle is repeated between the pre-configured minimum and maximum temperature periods up until the device fails. For the sake of the system's safety, a threshold temperature is set to halt the experiment via the protection relay in case the device cannot turn off. It is important to emphasize that the temperature has a strong influence on the value of $R_{DS(on)}$ -the higher the temperature, the higher the $R_{DS(on)}$. Hence, to eliminate that temperature effect, while monitoring the lifespan of the device, the data used for the RUL estimation model are collected once the device reaches the pre-configured minimum temperature. After the device temperature hits the minimum temperature, the device is turned on for $100\mu s$ and the data is collected with the $1kHz/100\mu s$ sampling rate. Due to the experimental setup limitation, noises and measurement problems could not be eliminated in this setup. The case temperature is assumed to be close enough to the junction temperature; thus, in this work, the device temperature refers to the case temperature. The conducted ALT experiment procedure is shown in Figure 7, and the experimental setup conditions are given in Table 10 in the Appendix.

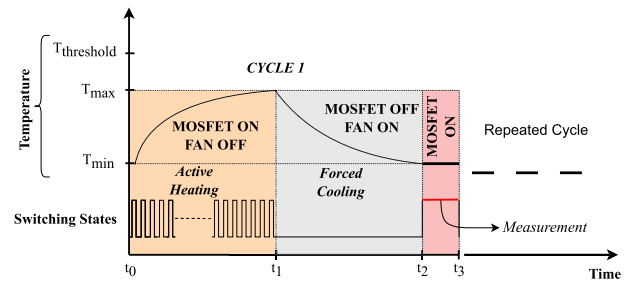


FIGURE 7. The operation method of the conducted experimental tests.

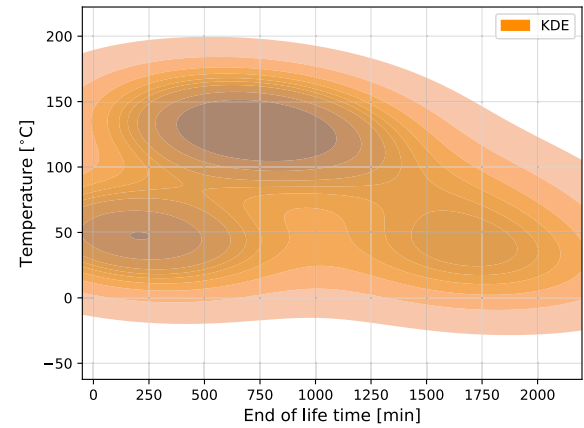


FIGURE 8. KDE of ALT measured temperature vs survival time to EoL.

V. EXPERIMENTAL DATA ANALYSIS

A. ALT DATA ANALYSIS

For the dataset creation, a total of 20 power MOSFET device samples were tested. All tested devices failed with gate loss meaning that the DUT cannot turn-off just before the complete failure. The change of the on-state resistance in each and every DUT increased beyond the permissible threshold 0.05Ω . However, for the sake of mission-critical application safety, the results in this paper only cover the increased 5% on-state resistance part. Figure 9 illustrates the distribution of initial MOSFET resistance when compared to MOSFET survival time, whilst Figure 8 illustrates cell survival time as a function of temperature. It is clear from Figure 9 that despite MOSFET's being from the manufacturer batch their initial resistance value is widely spread. As a consequence, traditional threshold methods of determining MOSFET SoH or RUL would result in erroneous predictions, hence the need for a dynamic adaptable approach such as the one proposed in this work.

1) MOSFET FAILURE ANALYSIS

Throughout the power cycle test, the module generates a significant temperature gradient which induces failures in time. During the temperature swings, the various coefficients of thermal expansion of materials and the resulting vertical and lateral temperature gradients cause mechanical stress at the interfaces. After the failure of DUTs, some are selected

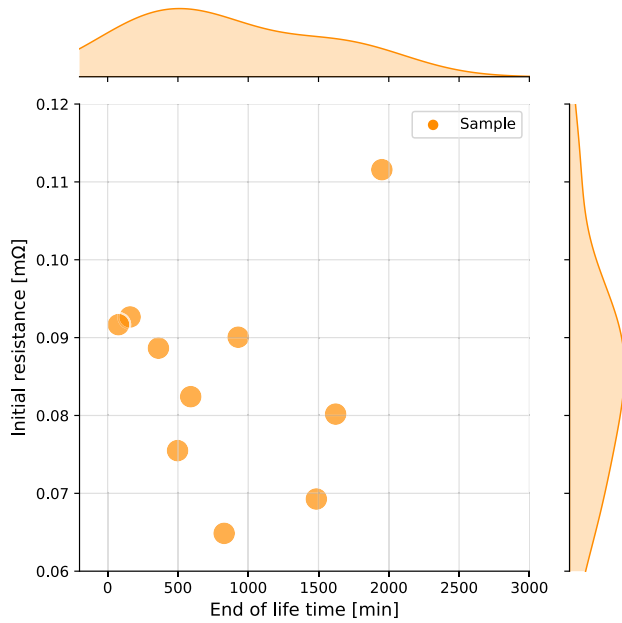


FIGURE 9. KDE of pre-ALT measured Resistance vs survival time to EoL.

for decapsulation and failure analysis using scanning electron microscopy (SEM). A destructive chemical decapsulation method (60% fuming nitric and 40% sulphuric) is applied to the devices to be able to observe the internal components of the encapsulated device. The tested device-5 failed as an open circuit. The SEM scanning of the sample shows the lift-off on the source pad as demonstrated in Figure 10. The adherence between the bond wire and on-the-chip metallization disappeared due to the cycling stress. Similarly, the reconstruction of the topside chip metallization is a process observed during active power cycles due to the significant temperature variation, shown in Figure 11. This contact metallization is generally composed of a grain-structured, vacuum-metallized aluminium layer. Due to the mismatch in thermal expansion between silicon and aluminium, this layer is subjected to significant strain during repeated temperature fluctuation. The reconstruction of the chip's metallization decreases the contact layer's density, hence increasing the contact's specific resistivity. Another failure mode observed after the ALT experiments is the bond wire heel cracking. It is due to the thermo-mechanical stress on the wires. Figure 12 (a-b) shows device 12 and device-15, which have heel crackings.

VI. RESULTS

As explained in Section III, the machine learning pipeline works in a hybrid fashion combining the classification and regression algorithms for SoH estimation and RUL prediction. This section provides the results of the proposed ML pipeline when all train-test samples are considered. Additionally, to simulate a real-time deployment and illustrate RF classifier SoH estimation and BRR forecasting of RUL, we randomly select Si MOSFET device no. 10. Please

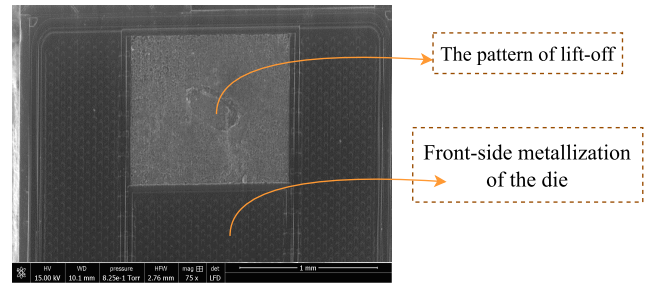


FIGURE 10. The lift-off failure mode observed on the source pad of Si-device 5.

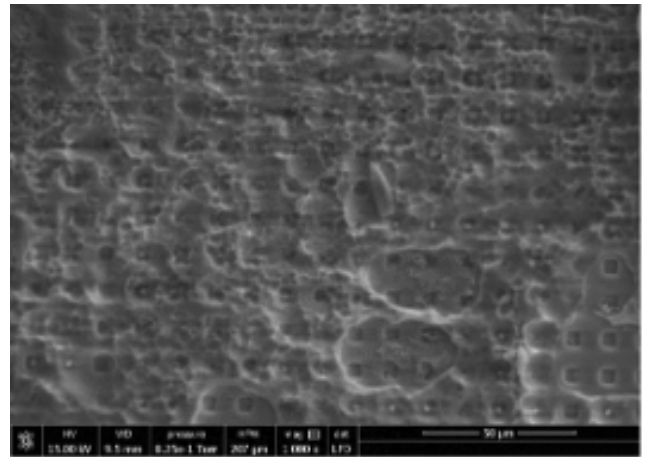


FIGURE 11. The reconstructed source-pad metallization after device fail.

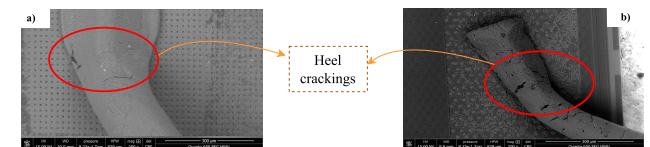


FIGURE 12. The bond wire crackings observed on the source pad of a) Device 12 and b) Device 15.

note that out of a total of 20 samples, 10 samples are used as validation data for classifier hyperparameter tuning and calibration, whilst the remaining data of 10 samples are used as train/test. To simulate a real-time deployment in evaluating the ML pipeline performance we iteratively trained on 9 samples and tested on a single sample until we ran out of samples (similar to cross-validation). We choose to assess the pipeline this way due to the limited dataset of just 10 samples (refer to Table 9 in the Appendix for details on which samples are used for validation, train and test).

A. STATE OF HEALTH ESTIMATION RESULTS

Subject to the training of the classifier that is part of the proposed ML pipeline as per Section III-B, we obtain the performance summarised in Table 2 and Table 3. The aforementioned tables summarise the results per class (health, pre-failure) as well as macro averages and weighted averages

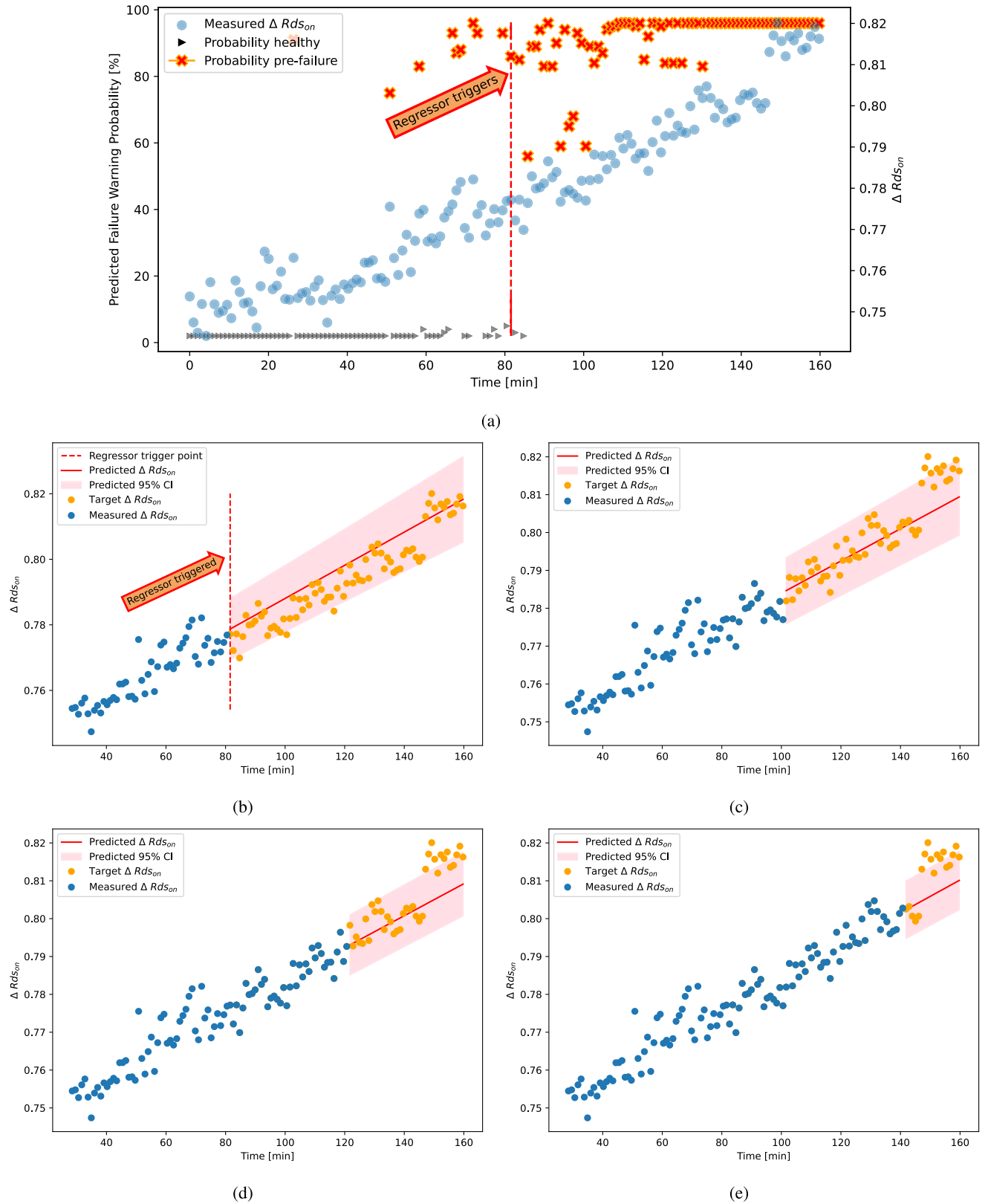


FIGURE 13. Results for S_1 sample no. 10. a RF Classifier predictions and regressor trigger point, **b** Linear BRR regressor RUL forecasting immediately after trigger point (100% to EoL), **c** Linear BRR regressor RUL forecasting 75% to EoL, **d** Linear BRR regressor RUL forecasting 50% to EoL, **e** Linear BRR regressor RUL forecasting 100% to EoL.

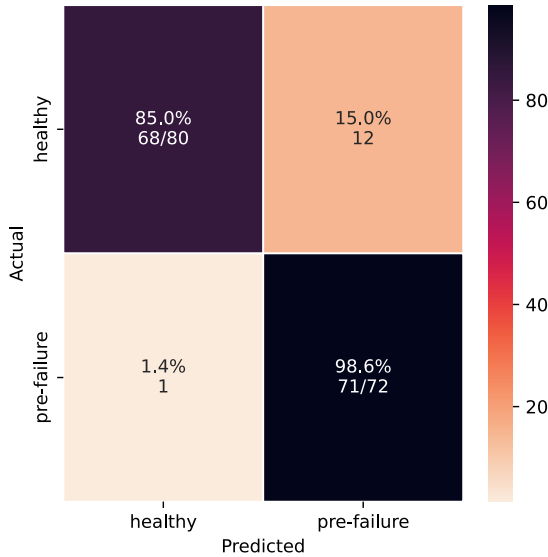


FIGURE 14. Confusion matrix for RF classifier when deployed on test sample Si MOSFET device no. 10.

of precision, recall, F1-score, and accuracy for all samples using different ML models (LGR, SVC) (refer to Table 8 in the Appendix for a detailed description of the metrics). The distribution between the class labels is illustrated in Appendix Figure 17, where we highlight that the ratio of healthy to pre-failure i.e. majority-to-minority ratio is 1.36. In other words, we are dealing with a low to moderate level of class imbalance. However, for transparency, we provide detailed metrics for macro and weighted averages. The weighted average takes into account the class imbalance by giving more importance to classes with a larger number of samples, whilst macro-averaging calculates the scores by treating all classes equally, regardless of the number of samples in each class. Finally, we compute macro accuracy, which, given the low/moderate class imbalance is indicative of the model's performance. The scores computed in Table 2 and Table 3 also provide the standard deviation of the model when assessed on each MOSFET sample. Overall, the RF model achieves an 80.7% accuracy with a standard deviation of $\pm 17\%$. We believe the high variance occurs due to some samples triggering the pre-failure early due to their short survival time.

To better illustrate the RF model's decision-making process we also provide under-the-hood visualisation of the model decision-making in Figure 13a when Si MOSFET sample no.10 is considered as the test sample (all other 9 samples are used for training of the classifier). In fig. 13a one can identify that the model is highly confident in classifying samples as healthy/pre-failure due to the output probability being close to 0 for healthy and close to 1 for pre-failure. The same figure also illustrates the identification of the regressor trigger point, currently defined as three consecutive pre-failure probabilities bigger than the optimum decision threshold determined using MCC (refer to Section III-B for MCC optimum decision threshold). When

TABLE 2. Overall different ML classifiers results.

	ML Model	Precision [%]	Recall [%]	F1-score
Healthy	Random Forest	81.4 $\pm$ 21	86.1 $\pm$ 19	82.2 $\pm$ 18
	Logistic Regression Classifier	74.0 $\pm$ 17	91.9 $\pm$ 7	80.6 $\pm$ 10
	Support Vector Classifier	83.3 $\pm$ 16	90.4 $\pm$ 8	85.0 $\pm$ 9
Pre-failure	Random Forest	84.9 $\pm$ 16	68.8 $\pm$ 34	67.4 $\pm$ 29
	Logistic Regression Classifier	83.0 $\pm$ 15	42.7 $\pm$ 33	48.1 $\pm$ 29
	Support Vector Classifier	85.6 $\pm$ 14	68.5 $\pm$ 34	66.8 $\pm$ 29

TABLE 3. RF classifier accuracy results.

	ML Model	Precision [%]	Recall [%]	F1-score
Accuracy [%]	Random Forest	-	-	80.7 $\pm$ 17
	Logistic Regression Classifier	-	-	73.5 $\pm$ 15
	Support Vector Classifier	-	-	80.8 $\pm$ 13
macro Average	Random Forest	83.2 $\pm$ 14	77.4 $\pm$ 18	74.8 $\pm$ 21
	Logistic Regression Classifier	78.5 $\pm$ 10	67.2 $\pm$ 14	64.3 $\pm$ 18
	Support Vector Classifier	84.4 $\pm$ 6	79.4 $\pm$ 14	76.0 $\pm$ 18
weighted Average	Random Forest	84.1 $\pm$ 14	79.7 $\pm$ 17	77.2 $\pm$ 20
	Logistic Regression Classifier	79.2 $\pm$ 9	73.6 $\pm$ 14	69.0 $\pm$ 19
	Support Vector Classifier	86.2 $\pm$ 6	80.8 $\pm$ 13	76.0 $\pm$ 18

TABLE 4. RF classifier results for Si MOSFET sample no. 10. Refer to Figure 13 for a visualisation of the predicted probabilities of the classifier.

	Precision [%]	Recall [%]	F1-score	No. of data points
Healthy	99	85	0.91	80
Pre-failure	86	99	0.92	
				Total no. of data points
Accuracy [%]	-	-	91	152
macro Average	82	92	0.91	152
weighted Average	92	91	0.91	152

deployed on the aforementioned sample, the RF classifier achieves an accuracy of 91% as per Table 4 and a weighted F1-score of 0.91 despite MOSFET Si no.10 sample having

TABLE 5. Overall Bayesian Ridge regression with both linear and 2nd order polynomial feature transformation. The RMSPE is calculated at 100%, 75%, 50%, and 25% to EoL. Refer to Figure 13 for an illustration of the intervals.

	RMSPE $\pm$ Var [%]				
	100% to EoL	75% to EoL	50% to EoL	25% to EoL	Average
BRR + linear	1.51 $\pm$ 0.88	1.31 $\pm$ 0.49	1.10 $\pm$ 0.41	1.07 $\pm$ 0.56	1.25 $\pm$ 0.48
BRR + 2 nd order poly	5.52 $\pm$ 7.03	3.76 $\pm$ 3.04	1.38 $\pm$ 0.79	1.20 $\pm$ 0.78	2.96 $\pm$ 2.25
RANSAC Regressor	4.71 $\pm$ 7.85	1.57 $\pm$ 0.50	1.40 $\pm$ 0.72	1.00 $\pm$ 0.56	2.17 $\pm$ 2.20

TABLE 6. Bayesian Ridge regression results for Si MOSFET sample no. 10. Refer to Figure 13 for a visualisation of the intervals associated with the results.

	RMSPE $\pm$ Var [%]				
	100% to EoL	75% to EoL	50% to EoL	25% to EoL	Average $\pm$ Var [%]
BRR + linear	0.68	1.16	1.02	1.06	0.98 $\pm$ 0.05
BRR + 2 nd order poly	5.76	3.14	0.86	0.81	2.64 $\pm$ 5.50
RANSAC Regressor	2.79	1.97	1.26	1.12	1.78 $\pm$ 0.6

a medium to high-class imbalance with majority-to-minority (healthy-to-pre-failure) class ratio of 2.47. In addition, we provide the confusion matrix for this sample in Figure 14. From Table 4 and Figure 14 one can identify that, in this case, the RF identifies 8 pre-failure data points as healthy. This implies that the RUL regressor is triggered slightly late for the MOSFET Si device no.10, however as per Figure 13 and later explained in the below section, this is sufficient to accurately forecast the RUL degradation curve and subsequently EoL of the device. Finally, once the classifier identifies the sample to have entered the pre-failure warning zone three consecutive times, the RUL regressor is triggered (see Figure 13a and Figure 13b).

B. REMAINING USEFUL LIFE PREDICTION RESULTS

To forecast degradation and thus predict the RUL, different regression models are employed, as detailed in Section III-C. The RUL is triggered once the classifier starts estimating the health of the MOSFET device entering pre-failure three consecutive times. The results of the BRR regression with both linear and polynomial feature transformation, and RANSAC regression are captured in Table 5.

We compute polynomial features by simply applying a polynomial transform on input data to the BRR algorithm, in this case, we use the time vector in minutes. We then fit the model on these features to obtain a non-linear model. Table 5 clearly indicates that the linear BRR model outperforms BRR with a 2nd order polynomial feature space on every interval to EoL as well as overall, achieving a lower variance. This is to be expected since most degradation curves (refer to Figure 6) exhibit a linear increase in resistance past the pre-failure decision threshold of 3%. To demonstrate the effectiveness of

our strategy and thus support timely maintenance scheduling decisions we illustrate in Figure 13 the inner-working of the RUL forecaster. The Figure 13 not only illustrates the forecasts of RUL, but also the confidence interval (CI) of the forecasts for the Si MOSFET sample no. 10 at four different stages of its life i.e once classifier triggered (100% to EoL), and at 75%, 50% and 25% to EoL respectively. As can be seen in Figure 13, the CI captures well the variance in the measured resistance all the way to the end of life of the device. The performance on the Si MOSFET sample no. 10 is captured in Table 6, whilst the average results are summarised in Table 5 demonstrating a similar performance on other test data. In conclusion, according to Table 6 the same performance trend is observed, where the linear model outperforms the 2nd order polynomial by a high margin in both RMSPE and its variance.

VII. DISCUSSION ON PRACTICAL APPLICABILITY OF THE PIPELINE

Currently, we are witnessing the 4th industrial revolution through intelligent automation and data collection fueled by the use of cyber-physical systems (CPS), the Internet of Things (IoT), smart sensors, and artificial intelligence (AI) [46]. The amount of operational data collected at components and sub-component levels is exponentially increasing, making it easier than ever before to deploy data-driven PHM methods for condition-based monitoring and RUL forecasting [47]. Data-driven PHM techniques such as the proposed ML pipeline can therefore be deployed in a data-rich setting, such as industrial manufacturing plants or critical infrastructure facilities where component operational data is recorded. In the case of power electronics, leading manufacturers like Siemens, General Electric, and ABB are already monitoring voltage and current vs. time signals for IGBTs and MOSFETs [48], [49]. Therefore, the proposed PHM pipeline can be used in critical power electronics applications where mitigating the risks associated with hazardous failures and optimizing replacement costs are priorities.

In recent years, hybrid models combining various data-driven techniques have shown significant promise in RUL estimation tasks. One notable example is the work where a hybrid deep learning model for RUL estimation of reed relays was proposed [50]. This model integrates dynamic time warping-based K-means clustering to distinguish different degradation patterns and utilizes a combination of convolutional neural networks (CNN) and long short-term memory networks (LSTM) to handle the temporal correlations in the data. The results demonstrated superior performance compared to traditional CNN and LSTM models alone. While our approach similarly combines multiple models as in [50], our pipeline leverages different machine learning techniques (Random Forest and Bayesian Ridge Regression) and focuses on the specific challenges associated with power electronics, offering unique advantages in terms of computational efficiency and adaptability to different

devices and degradation behaviours. Our ML pipeline makes use of widely used algorithms in the PHM industry, such as Random Forest and BRR [51], [52]. In addition, no work to date explores the idea of SoH estimation of power electronics when there is no operational data available from the beginning of the component's life. This is thus a considerable problem for companies looking to deploy machine learning for maintenance strategies of power electronics only when snapshots of operational data are available. For this reason, our work uses operational data at any point in time in the component's life cycle to first estimate the health of the component without the need for historical data for that particular component. Furthermore, the pipeline only triggers the RUL prediction once the component's SoH indicator reaches a pre-failure threshold, in this case, the resistance increased by 3% from the pristine condition. Therefore, the estimation of SoH and subsequent integration of SoH into RUL prediction into a single state-alone model represents significant progress in the field of prognostics and health management, and it is a major contribution of the work. By combining SoH and RUL into a unified framework, our model allows for continuous monitoring and forecasting of the system's state. The simultaneous evaluation of SoH and RUL also facilitates proactive and well-informed decision-making in maintenance procedures. Furthermore, to facilitate the deployment of the proposed pipeline both in the cloud and on the edge, such as on Nvidia's edge computing devices, we also reduce computational time and resources by first estimating device SoH using the RF algorithm and only triggering the BRR when the device is classified as pre-failure by the RF, forecasting its resistance degradation trajectory as explained above. Therefore, we simultaneously reduce computational resources and storage in comparison to the seminal works in [33] and [53] that require historical data from the beginning of the component's life-cycle and attempt to predict RUL as early as the first few minutes of component operation when the component is known to be new.

The proposed machine learning model was validated on power MOSFETs operated under realistic use cases since these devices are susceptible to failure when onboard numerous devices, including EVs, traction, and aerospace applications [54]. During the operation of the power MOSFETs, we recorded the voltage and current signals typical of industrial practices. We then applied standard machine learning feature engineering techniques based on statistical analysis, as seen in other seminal works [55]. Whilst conventional approaches [29], [56] rely on window analysis and averaging, they also require operational data from the beginning of the device's life cycle. In comparison, our pipeline discards the data once the SoH assessment has been made. We only save data when the model enters pre-failure mode which therefore requires us to forecast its RUL. We have chosen BRR as the algorithm to forecast RUL because it is less computationally demanding than the

Kalman filters used in [25] or Long Short-Term Memory (LSTM) neural networks used in [33]. Kalman filters are often used in more complex, dynamic models, which can be more computationally demanding [57], while LSTMs neural networks have been developed for applications where the order of data points is critical, like natural language processing, time series analysis, and sequence generation [58]. This implies that LSTMs require a robust historical dataset and are generally more complex, with a larger number of parameters to train when compared to a BRR [59]. As illustrated in Figure 13 degradation of resistance beyond a 3% increase in the SoH indicator (resistance) is adequately captured by the linear BRR. In fact, Table 5 also illustrates results for a BRR with a 2nd-order polynomial, where it is clearly seen that a non-linear RUL model achieves a higher RMSPE of 1.67% when compared to a linear BRR of 0.79%. This indicates that degradation beyond 80% SoH is linear and therefore complex algorithms such as LSTM neural networks are not needed. In addition, we provide both the data and code as open-source for the research community to validate our work (please follow [60] for the data and code). Whilst our models achieve high accuracy, we believe future work must include investigations of temperature effects on the SoH estimation model. Temperature variations during operation could further introduce uncertainty into the measurement of the voltage and current signals and thus propagate such errors into the estimation algorithm. Possible mitigation strategies include the use of temperature as an input when training the SoH models so that the algorithm learns the correlation between temperature, generated features, and the SoH indicator. Additionally, further experimentation in the data processing stage should be carried out, and an analysis of the window size (see Section III-A1 for an explanation of signal processing) effect on feature engineering and subsequently on model accuracy must be investigated. In summary, this work proposes a framework for the integration of SoH estimation and RUL prediction and can therefore be used for other components, including capacitors or power semiconductors such as IGBTs, however, more work is required in the future to empirically prove its usefulness when applied to other components.

VIII. CONCLUSION

The degradation of electronic devices over time poses a significant challenge and cost our economy an estimated 53.6 million tonnes of e-waste in 2019 [61], making it crucial to accurately predict their remaining useful lives. To date, corrective and preventive maintenance are the most common strategies in the industry to prevent catastrophic failures. However, these methods are neither reliable nor cost-effective. Owing to improvements in condition-monitoring tools, it is increasingly becoming common practice to collect critical device parameters and make use of data-driven techniques to infer and forecast devices' health. Therefore, this study proposes a novel sequential two-step model

that leverages machine learning techniques to estimate the device's SoH and subsequently forecast the remaining useful life of power MOSFETs once the device enters a pre-failure state. Specifically, the proposed machine learning pipeline utilises a Random Forest classifier for estimating the device SoH, measured as the percent increase in resistance, and a Bayesian-Ridge Regression for RUL forecasting. To demonstrate the pipeline's effectiveness, it was deployed on power MOSFET degradation data obtained through an accelerated ageing test setup. The results obtained from the experiments indicate that the proposed hybrid model is capable of accurately predicting the SoH with 80% accuracy and forecasting the remaining useful life of a power MOSFET with an RMSPE of 1.25%. Finally, given the sequential architecture of the proposed approach that triggers the RUL forecasting only when needed, we also demonstrated it is computationally efficient, making it a practical solution for real-time applications. The successful implementation of this sequential two-step model represents a significant step forward in the field of PHM for electronic devices exhibiting a non-linear degradation behaviour. By accurately forecasting the remaining useful life, organisations can optimise maintenance schedules, reduce downtime, and make informed decisions regarding device replacement. This study contributed to enhancing the reliability and performance of electronic systems, ultimately benefiting various industries and applications. Future research may explore additional machine learning techniques and expand the scope to different types of electronic devices for further improvements in predictive maintenance.

APPENDICES

A. MOSFET STRUCTURE

The total on-state resistance of MOSFET with trench technology ($R_{ds(on)_trench}$) can be obtained as follows eq. (4). In these equations, R_{CD} and R_{CS} are the resistances of drain and source contacts, respectively. The channel resistance and N+ source region are denoted as R_{Ch} , R_{N+} . R_A , R_D and R_{SUB} are the resistances of accumulation, drift-region, and substrate, respectively [11]:

$$R_{ds(on)_trench} = R_{CD} + R_{CS} + R_{Ch} + R_{N+} + R_A + R_D + R_{SUB} \quad (4)$$

B. PERFORMANCE METRICS

Every machine learning model, whether it is linear regression or classification, requires a performance metric. There are many different performance metrics used in different ML algorithms. However, in this section, the metrics incorporated in the developed RUL prediction model are given in the following.

a) Classification Metrics:

Accuracy is the number of correct predictions (true positive and true negative) divided by the total number of predictions multiplied by 100. Accuracy is possibly the

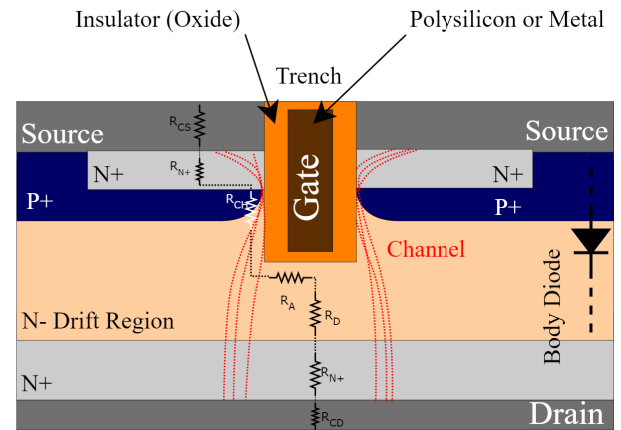


FIGURE 15. Trench Technology MOSFET Structure.

TABLE 7. The experimental design for power MOSFET ALTs.

Parameter	Value
Si MOSFET Device	IRF510 (TO-220) [62]
Data Acquisition and Controller	NI cDAQ-9178 [63]
Thermocouple Module	NI-9213 (Sample Rate: 75S/s) [64]
Analog Input Module	NI-9221 (Max Input: $\pm 60V$, Sample Rate: 800kS/s, Error Rate: $\pm 0.07\%$) [65]
Digital Output Module	NI-9475 (Max Output: 60V, Update Rate: 1 μ s) [66]
Protection Relay	Multicomp Pro Power Relay (Nominal Voltage: 12V)
Current Sensor	LEM Transducer (Nominal Current: 25A, Accuracy: 0.8%)
IR Temperature Sensor	CTlaser LT (Response Time: 9ms)
Load Resistor	Vishay 0.5 Ω and 5 Ω [67]

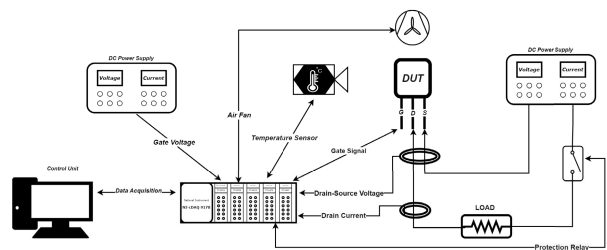


FIGURE 16. The general overview of the experimental test setup.

simplest measure to use and execute and can be expressed as follows eq. (5).

$$accuracy = \frac{tp + tn}{tp + tn + fp + fn} * 100\% \quad (5)$$

Here, true positive (tp) is the number of positive class samples, and true negative (tn) is the number of negative class samples that your model properly predicted. False positive (fp) is the number of negative class samples, and false negative (fn) is the number of positive class samples that your model incorrectly predicted.

TABLE 8. Time-domain statistical characteristic parameters.

Features	Formulas
Mean	$\mu = \frac{1}{n} \sum_{t=1}^n x_t$
Standard Deviation	$\text{std} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (x_t - \mu)^2}$
Skewness	$\text{skn} = \frac{1}{n} \sum_{t=1}^n (x_t - \mu)^3$
Kurtosis	$\text{krt} = \frac{1}{n} \sum_{t=1}^n (x_t - \mu)^4$

TABLE 9. Train, validation, test data set split.

MOSFET type	Sample no.	Total no. of Sampels	Data type
Si	1, 2, 4, 5, 8, 9, 17, 19, 20, 22, 30	11	Validation
Si	3, 10, 12, 15, 16, 18, 25, 28, 29	9	Train/Test
Total no. of validation samples		11	
Total no. of train/test samples		9	

Precision is the ratio between true positives and predicted total positives, meaning that it gives the accuracy of the positive predictions as shown in eq. (6).

$$\text{precision} = \frac{tp}{tp + fp} * 100\% \quad (6)$$

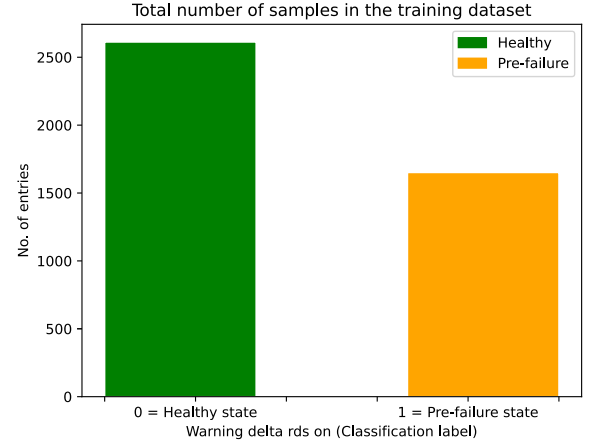
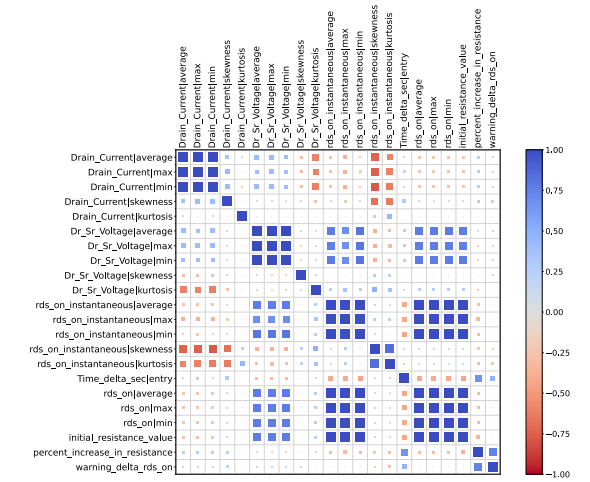
Recall or **sensitivity** is the ratio of true positives to all positives in the ground truth. It can be expressed as following eq. (7):

$$\text{recall} = \frac{tp}{tp + fn} \quad (7)$$

F1-score incorporates both precision and recall. The F1 score is actually the harmonic mean of the two scores. It can be expressed as following eq. (8). The harmonic mean put a considerably greater emphasis on low values. Consequently, the classifier will only receive a high F1 score if recall and precision are both high.

$$F1 - \text{score} = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} = \frac{2 * tp}{2 * tp + fp + fn} \quad (8)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \quad (9)$$

**FIGURE 17.** Overall class imbalance between the health/pre-failure classes in the training data used for the classifier.**FIGURE 18.** Correlation matrix using Pearson's correlation coefficient of all features in the training data.

where *TP* (*True Positives*) represents the number of correctly predicted positive instances, *TN* (*True Negatives*) represents the number of correctly predicted negative instances, *FP* (*False Positives*) represents the number of incorrectly predicted positive instances, *FN* (*False Negatives*) represents the number of incorrectly predicted negative instances.

b) Regression Metrics:

Root Mean Squared Error (RMSE) is a common metric used in regression problems for measuring the error of a model. It gives an estimate of the typical error rate of the system's predictions, with a higher weight for significant mistakes [68]. Equation (10) illustrates the formula for calculating the RMSE:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - y_i^*)^2} \quad (10)$$

where y_i is the ground truth value, while y_i^* is the predicted value from the regression model and N is the number of data points.

TABLE 10. Design of experiment.

Device Type	Device Number	Min. Temp.	Max. Temp.	Delta T	Switching Frequency	Duty Cycle	Gate Voltage	Drain Voltage	Load Resistance
Si-MOSFETs	1, 2, 3, 4, 5	50°C	170°C	120°C	2kHz	50%	12V	10V	10hm
	8, 9	45°C	165°C	120°C	10kHz	50%	12V	10V	10hm
	10, 12	45°C	165°C	120°C	2kHz	50%	12V	10V	10hm
	15, 16, 17	130°C	250°C	120°C	10kHz	50%	12V	10V	20hm
	18, 19, 20	130°C	250°C	120°C	2kHz	50%	12V	10V	20hm
	29	30°C	165°C	135°C	10kHz	50%	12V	10V	0.50hm
	28, 30	30°C	165°C	135°C	10kHz	50%	12V	10V	0.50hm
	22	80°C	240°C	160°C	10kHz	50%	12V	10V	20hm
	25	80°C	240°C	160°C	2kHz	50%	12V	10V	20hm

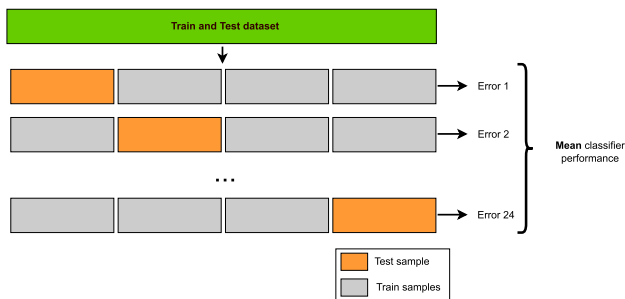


FIGURE 19. Train - test sample split and classifier performance evaluation method.

Root Mean Squared Percentage Error is another loss metric used in regression problems. It estimates the average deviation of a test or cross-validation observation from its predicted value. Equation (11) illustrates the mathematical expression for calculating the RMSPE:

$$RMSPE = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{y_i - y_i^*}{y_i^*} \right)^2} * 100\% \quad (11)$$

CODE AND DATA AVAILABILITY

The datasets used in this study and the code developed for this work are available: GitHub Repository

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